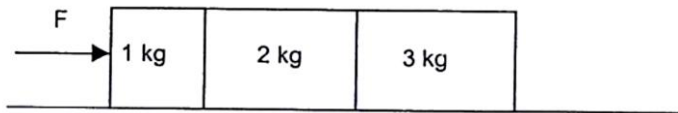


$$t_2 - t_1$$

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Three boxes are pushed along a smooth surface by a force F as shown below. What is the force exerted by the 2 kg mass on the 1 kg mass?



A

$$\frac{1}{3} F$$

B

$$\frac{1}{2} F$$

C

$$\frac{3}{4} F$$

D

$$\frac{5}{6} F$$

Ans: D

acceleration of boxes = $F/6$. Net force on 1 kg box = $1 \times F/6 = F/6$.

Net force on 1 kg box = $F - \text{contact force by 2kg} = F/6$

Contact force by 2kg = $5/6 F$

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